22
Symbolic integration

The basic task in this chapter is, given an "expression" f, say $f \in F(x)$, where F is a field, to compute the indefinite integral $\int f = \int f(x)dx$, that is, another "expression" (possibly in a larger domain) g with $g' = f$, where $'$ denotes differentiation with respect to the variable x. "Expressions" are usually built from rational functions and "elementary functions" such as sin, cos, exp, log, etc. (Since it is more common, we denote the natural (base e) logarithm by "log" instead of "ln" in this chapter.) Such integrals need not exist: Liouville's (1835) theorem implies that $\exp(x^2)$ has no integral involving only rational functions, sin, cos, exp, and log.

A practical approach to the symbolic integration problem is to use a plethora of formulas for special functions, tricks from basic calculus like substitutions and integration by parts, and table lookups. There are projects that load the whole contents of existing printed integral tables into computer algebra systems, using optical character recognition, and modern computer algebra systems can solve practically all integration exercises in calculus textbooks. In the following, we discuss a systematic algorithm in the case of rational and "hyperexponential" functions as integrands. This approach can be extended—with plenty of new ideas and techniques—to more general functions, but we do not pursue this topic further.

22.1. Differential algebra

A remarkable fact about symbolic integration is that the theory can be set up in a purely algebraic way, without any use of limit processes. We are familiar with this approach from Chapter 9, where we used a formal Taylor expansion for Newton iteration.

DEFINITION 22.1. *We let R be an integral domain of characteristic zero, and $D: R \longrightarrow R$ such that the following hold for all $f, g \in R$.*

(i) $D(f+g) = D(f) + D(g)$,

(ii) $D(fg) = D(f)g + fD(g)$ *(Leibniz rule).*

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Then D is called a ***differential operator*** *(or derivation, or derivative), and (R,D) is a* ***differential algebra*** *(or* ***differential field****, if R is in fact a field).*

The set $R_0 = \{c \in R : D(c) = 0\}$ is the ***ring of constants*** *of (R,D). If $f,g \in R$ are such that $D(f) = g$, then we say that f is an* ***integral*** *of g and write $f = \int g$.*

The notation $f = \int g$ is merely a relation and not a function since, as in calculus, we may add an arbitrary constant $c \in R_0$ to f to get another integral of g.

LEMMA 22.2. *In a differential algebra (R,D), the usual properties hold for all $f,g \in R$:*

(i) $D(1) = 0$,

(ii) D *is R_0–linear:* $D(af + bg) = aD(f) + bD(g)$ *for* $a,b \in R_0$,

(iii) $D\left(\dfrac{f}{g}\right) = \dfrac{D(f)g - fD(g)}{g^2}$ *if g is a unit,*

(iv) $D(f^n) = nf^{n-1}D(f)$ *for* $n \geq 1$,

(v) $\int (fD(g)) = fg - \int (D(f)g)$ *(integration by parts).*

See Exercise 22.2 for a proof.

EXAMPLE 22.3. (i) $D(a) = 0$ for all $a \in R$. This is the **trivial derivative** on R, with $R_0 = R$.

(ii) $R = \mathbb{Q}(x)$, $D(x) = 1$, and $D(a) = 0$ for all $a \in \mathbb{Q}$. This gives the usual derivative: $D(\sum_i f_i x^i) = \sum_i i f_i x^{i-1}$ when all f_i are rational numbers. Here, $R_0 = \mathbb{Q}$ (Exercise 22.4), and polynomials are easily integrated as

$$\int \sum_i f_i x^i = \sum_i \frac{f_i}{i+1} x^{i+1}. \diamond$$

In what follows, we write f' instead of $D(f)$ for the usual derivative on $F(x)$, where F is a field of characteristic zero.

LEMMA 22.4. *The rational function $1/x \in \mathbb{Q}(x)$ has no rational integral:*

$$\forall f \in \mathbb{Q}(x) \quad f' \neq \frac{1}{x}.$$

See Exercise 22.5 for a proof. The lemma motivates the need for domain extensions when looking for integrals: the usual derivation on $\mathbb{Q}(x)$ is not surjective, so that we need logarithms.

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DEFINITION 22.5. *Let (L,D) be a differential field, $K \subseteq L$ a subfield, $f \in L$, and $u \in K$ with $D(f) = D(u)/u$. Then we write $f = \log u$ and say that f is* ***logarithmic over K****.*

We note that log is in general a relation, not a function, since we may add an arbitrary constant to f and get another integral of $D(u)/u$.

EXAMPLE 22.6. (i) We let $u = x \in K = \mathbb{Q}(x)$, $L = \mathbb{Q}(x,f)$, where f is a new indeterminate, and $f' = u'/u = 1/x$, so that $f = \log x$. Then $\displaystyle\int \frac{1}{x} = \log x$.

(ii) Similarly, $\displaystyle\int \frac{1}{x^2-2} = \frac{\sqrt{2}}{4}\log(x-\sqrt{2}) - \frac{\sqrt{2}}{4}\log(x+\sqrt{2})$ in the differential field $\mathbb{Q}(\sqrt{2})(x,\log(x-\sqrt{2}),\log(x+\sqrt{2}))$.

(iii) $$\int \frac{1}{x^3+x} = \log x - \frac{1}{2}\log(x+i) - \frac{1}{2}\log(x-i) = \log x - \frac{1}{2}\log((x+i)(x-i))$$
$$= \log x - \frac{1}{2}\log(x^2+1) \in \mathbb{Q}(x,\log x,\log(x^2+1)).\ \diamond$$

22.2. Hermite's method

Let F be a field of characteristic zero, $f,g \in F[x]$ nonzero and coprime, and suppose that we want to compute $\int (f/g)$. The idea is to find first $a,b,c,d \in F[x]$ with

$$\int \frac{f}{g} = \frac{c}{d} + \int \frac{a}{b}, \tag{1}$$

$\deg a < \deg b$, and b squarefree. The rational function c/d is called the **rational part**, $\int (a/b)$ the **logarithmic part** of the integral; we deal with the latter in the next section.

First, the polynomial part is separated from f/g by performing one division with remainder. This yields $q,h \in F[x]$ with $f = qg + h$ and $\deg h < \deg g$, so that $f/g = q + h/g$, and h/g is a proper rational function. Integration of the polynomial q is of course trivial.

We compute the squarefree decomposition of g, that is, $g_1,\ldots,g_m \in F[x]$ monic, squarefree and pairwise coprime with $g_m \neq 1$ and $g = g_1 g_2^2 \cdots g_m^m$ (Section 14.6). Next, we calculate the corresponding partial fraction decomposition

$$\frac{h}{g} = \sum_{1\le i\le m} \sum_{1\le j\le i} \frac{h_{ij}}{g_i^j}, \tag{2}$$

with all $h_{ij} \in F[x]$ of smaller degree than g_i (see Section 5.11).

Now comes the Hermite reduction process. Let $i \in \{1,\ldots,m\}$. We reduce $\int h_{ij}/g_i^j$ to a sum of a rational function plus an integral of the same form, but with

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the index j lowered by 1, from $j = i$ down to $j = 2$. Using that $\gcd(g_i, g_i') = 1$ since g_i is squarefree, we can compute $s,t \in F[x]$ such that $sg_i + tg_i' = h_{ij}$ and $\deg s, \deg t < \deg g_i$, by Theorem 4.10. Then

$$\begin{aligned} \int \frac{h_{ij}}{g_i^j} &= \int \frac{s}{g_i^{j-1}} + \int \frac{t \cdot g_i'}{g_i^j} = \int \frac{s}{g_i^{j-1}} + \frac{-t}{(j-1)g_i^{j-1}} + \int \frac{t'}{(j-1)g_i^{j-1}} \\ &= \int \frac{s + t'/(j-1)}{g_i^{j-1}} + \frac{-t}{(j-1)g_i^{j-1}}, \end{aligned} \tag{3}$$

using integration by parts. Adding $s + t'/(j-1)$ to $h_{i,j-1}$ and proceeding recursively leads to the following theorem. We use the multiplication time M as described on the inside back cover.

THEOREM 22.7 *Hermite reduction.*
The problem of integrating rational functions over a field F of characteristic zero can be reduced to integrating rational functions with squarefree denominator in softly linear time. More specifically, for nonzero polynomials $f, g \in F[x]$ of degree at most n, we can compute polynomials $q, c, d, a_1, \ldots, a_m \in F[x]$ and squarefree and pairwise coprime polynomials $g_1, \ldots, g_m \in F[x]$ such that

$$\frac{f}{g} = q + \left(\frac{c}{d}\right)' + \frac{a_1}{g_1} + \cdots + \frac{a_m}{g_m},$$

$\deg q \le n$, $\deg c < \deg d < n$, $\deg a_i < \deg g_i$ for all i, and $\sum_{1 \le i \le m} \deg g_i < n$, using $O(\mathsf{M}(n)\log n)$ operations in F.

PROOF. Splitting off the polynomial part is one division with remainder, taking $O(\mathsf{M}(n))$ operations in F. The squarefree decomposition of g can be computed using $O(\mathsf{M}(n)\log n)$ operations, by Theorem 14.23. Using fast Chinese remaindering (Section 10.3), the partial fraction decomposition (2) can again be computed with $O(\mathsf{M}(n)\log n)$ arithmetic operations (Exercise 10.18). Let $d_i = \deg g_i$ for all i. To analyze the cost of Hermite reduction, let $1 \le j \le i \le m$. Then one Hermite step (3) takes $O(\mathsf{M}(d_i)\log d_i)$ operations for computing s and t, plus $O(d_i)$ operations for updating $h_{i,j-1}$, in total $O(i\mathsf{M}(d_i)\log d_i)$ operations per g_i. Now

$$\sum_{1 \le i \le m} i\mathsf{M}(d_i)\log d_i \le (\log n)\mathsf{M}\Big(\sum_{1 \le i \le m} id_i\Big) \le \mathsf{M}(n)\log n,$$

since $\sum_{1 \le i \le m} id_i = \deg g \le n$, and the claim follows. □

It is sufficient to use the fast Extended Euclidean Algorithm only once per g_i to compute $s^*, t^* \in F[x]$ such that $s^* g_i + t^* g_i' = 1$, and then for each j the polynomials s,t can be obtained from s^*, t^* and h_{ij} as described in Section 4.5, using only

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$O(\mathsf{M}(d_i))$ arithmetic operations. This does not affect the asymptotic time bound but is a practical improvement.

A different approach, due to Horowitz (1971), is the method of undetermined coefficients. Splitting off the polynomial part if necessary, we may assume that $\deg f < \deg g$. All denominators arising via (3) outside an integral divide the product $g_2 g_3^2 \cdots g_m^{m-1}$, and hence so does the denominator d in (1). Thus we may take the latter polynomial as d, the squarefree part (Section 14.6) of g as b, and plug a of degree $\deg b - 1$ and c of degree $\deg g - \deg b - 1$ with unknown coefficients into (1). This yields a linear system of equations for the coefficients of a and c, which has a coefficient matrix with at most n rows and columns, and it can be solved in time $O(n^3)$ using Gaussian elimination. Hermite reduction, however, is asymptotically faster by nearly two orders of magnitude.

22.3. The method of Lazard, Rioboo, Rothstein, and Trager

It remains to compute $\int (a/b)$ for nonzero coprime polynomials $a, b \in F[x]$ with b squarefree. Many calculus texts provide the following solution. Let K be the splitting field of b over F, that is, the smallest algebraic field extension of F containing all roots of b. Then $b = \prod_{1 \le k \le n} (x - \lambda_k)$, with distinct $\lambda_1, \ldots, \lambda_n \in K$. There exist $c_1, \ldots, c_n \in K$ such that $a/b = \sum_{1 \le k \le n} c_k/(x - \lambda_k)$ is the partial fraction decomposition of the rational function a/b, and the integral is

$$\int \frac{a}{b} = \sum_{1 \le k \le n} c_k \log(x - \lambda_k) \in K(x, \log(x - \lambda_1), \ldots, \log(x - \lambda_n)).$$

EXAMPLE 22.6 (continued). The integral in (iii) has been expressed without any algebraic extension of the field of constants $\mathbb{Q}$, while it is not clear how to do that with the integral in (ii). In fact, from what we will prove below, it is impossible to write the latter integral as a sum of logarithms with rational arguments. $\diamond$

This example shows that it may be unwise to compute the integral using the complete partial fraction decomposition. The following method computes the integral with as small an algebraic extension of the field of constants as possible.

THEOREM 22.8 *Rothstein's and Trager's integration algorithm.* *Let $a, b \in F[x]$ be coprime with $\deg a < \deg b = n$, and b monic and squarefree. If E is an algebraic extension of F, $c_1, \ldots, c_l \in E \setminus \{0\}$ are pairwise distinct, and $v_1, \ldots, v_l \in E[x] \setminus E$ are monic, squarefree, and pairwise coprime, then the following are equivalent:*

(i) $\displaystyle \int \frac{a}{b} = \sum_{1 \le i \le l} c_i \log v_i.$

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(ii) The polynomial $r = \operatorname{res}_x(b, a - yb') \in F[y]$ *splits over* E *in linear factors,* $c_1, \dots, c_l$ *are precisely the distinct roots of* r*, and* $v_i = \gcd(b, a - c_i b')$ *for* $1 \le i \le l$*. Here,* res_x *denotes the resultant with respect to the variable* x *(Chapter 6).*

PROOF. (i) $\Longrightarrow$ (ii): Differentiating (i) yields

$$\frac{a}{b} = \sum_{1 \le i \le l} c_i \frac{v_i'}{v_i},$$

or equivalently,

$$a \cdot \prod_{1 \le j \le l} v_j = b \cdot \sum_{1 \le i \le l} c_i u_i v_i',$$

where $u_i = \prod_{1 \le j \le l,\ j \ne i} v_j$. We claim that $b = \prod_{1 \le j \le l} v_j$ and $a = \sum_{1 \le i \le l} c_i u_i v_i'$.

Since a and b are coprime, b divides $\prod_{1 \le j \le l} v_j$. On the other hand, v_j divides $b \sum_{1 \le i \le l} c_i u_i v_i'$ and $v_j \mid u_i$ for $i \ne j$, whence $v_j \mid b \cdot c_j u_j v_j'$. But $c_j \in E$ is nonzero, $\gcd(v_j, u_j) = 1$, and $\gcd(v_j, v_j') = 1$, so that $v_j \mid b$ for $1 \le j \le l$. By the relative primality of the v_j, we also have $\prod_j v_j \mid b$. This implies that $b = \prod_{1 \le j \le l} v_j$, since b and all the v_j are monic, and $a = \sum c_i u_i v_i'$, as claimed.

Now Lemma 14.22 yields

$$\gcd(b, a - cb') = \begin{cases} v_j & \text{if } c = c_j \text{ for some } j \in \{1, \dots, l\}, \\ 1 & \text{otherwise,} \end{cases}$$

and for any $c \in E$

$$c \in \{c_1, \dots, c_l\} \iff \gcd(b, a - cb') \ne 1 \iff \operatorname{res}_x(b, a - cb') = 0 \iff r(c) = 0,$$

by Lemma 6.25. Thus r splits over E, and $\{c_1, \dots, c_l\}$ are precisely the distinct roots of r.

(ii) $\Longrightarrow$ (i): Let K be the splitting field of b over E, and $\lambda_1, \dots, \lambda_n \in K$ pairwise distinct with $b = \prod_{1 \le k \le n} (x - \lambda_k)$. Since b is squarefree, $b'(\lambda_k) \ne 0$ for $1 \le k \le n$. For $c \in K$, we have

$$\begin{aligned} r(c) = 0 &\iff \operatorname{res}_x(b, a - cb') = 0 \iff \gcd(b, a - cb') \ne 1 \\ &\iff \exists k \in \{1, \dots, n\} \quad (a - cb')(\lambda_k) = 0 \\ &\iff \exists k \in \{1, \dots, n\} \quad c = \frac{a(\lambda_k)}{b'(\lambda_k)}. \end{aligned}$$

We write $d_k = a(\lambda_k)/b'(\lambda_k) \in K$ for $1 \le k \le n$. Then every root c_j of r is equal to d_k for some $k \in \{1, \dots, n\}$, and conversely every d_k is a root of r, so that $\{c_1, \dots, c_l\} = \{d_1, \dots, d_n\}$. In particular, we have $l \le n$. The values $d_1, \dots, d_n$ need not be distinct.

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We now claim that v_j has precisely those λ_k as zeroes where $d_k = c_j$. Let $\sigma\colon\{1,\dots,n\} \longrightarrow \{1,\dots,l\}$ be such that $d_k = c_{\sigma(k)}$ for all k. Then σ is surjective,

$$(a - c_{\sigma(k)}b')(\lambda_k) = (a - d_k b')(\lambda_k) = 0$$

for $1 \le k \le n$, and $v_{\sigma(k)}(\lambda_k) = 0$ since $(x - \lambda_k)$ divides $\gcd(b, a - c_{\sigma(k)}b') = v_{\sigma(k)}$. Conversely, if $v_j(\lambda_k) = 0$ for some j, then also $(a - c_j b')(\lambda_k) = 0$, so that $c_j = d_k$ and $j = \sigma(k)$. Thus

$$v_j = \prod_{1\le k\le n,\ \sigma(k)=j} (x - \lambda_k),$$

and $b = \prod_{1\le j\le l} v_j$.

Now let $u_i = \prod_{1\le j\le l,\ j\ne i} v_j$ for $1 \le i \le l$, as before. Then

$$\begin{aligned}\Big(\sum_{1\le i\le l} c_i u_i v_i'\Big)(\lambda_k) &= (c_{\sigma(k)} u_{\sigma(k)} v'_{\sigma(k)})(\lambda_k) = c_{\sigma(k)}\Big(\sum_{1\le i\le l} u_i v_i'\Big)(\lambda_k)\\ &= d_k b'(\lambda_k) = a(\lambda_k)\end{aligned}$$

for $1 \le k \le n$. Both a and $\sum_{1\le i\le l} c_i u_i v_i'$ have degrees less than n and interpolate the same values at the n points $\lambda_1, \dots, \lambda_n$, and thus are equal. Hence

$$\Big(\sum_{1\le i\le l} c_i \log v_i\Big)' = \sum_{1\le i\le l} c_i \frac{v_i'}{v_i} = \frac{1}{b}\cdot \sum_{1\le i\le l} c_i u_i v_i' = \frac{a}{b}. \ \square$$

EXAMPLE 22.6 (continued). Let $a = 1$ and $b = x^3 + x$, as in Example 22.6 (iii). Then

$$r = \operatorname{res}_x(x^3 + x, 1 - y(3x^2+1)) = \det\begin{pmatrix} 1 & 0 & -3y & 0 & 0\\ 0 & 1 & 0 & -3y & 0\\ 1 & 0 & -y+1 & 0 & -3y\\ 0 & 1 & 0 & -y+1 & 0\\ 0 & 0 & 0 & 0 & -y+1\end{pmatrix}$$

$$= -4y^3 + 3y + 1 = -(2y+1)^2(y-1),$$

and its zeroes are $c_1 = -1/2$ and $c_2 = 1$. Thus

$$\begin{aligned} v_1 &= \gcd(x^3 + x, 1 + \frac{1}{2}(3x^2+1)) = \gcd\Big(x^3 + x, \frac{3}{2}(x^2+1)\Big) = x^2 + 1,\\ v_2 &= \gcd(x^3 + x, 1 - (3x^2+1)) = \gcd(x^3 + x, -3x^2) = x.\end{aligned}$$

Now $u_1 = b/v_1 = x$, $u_2 = b/v_2 = x^2 + 1$, and we check that $v_1 v_2 = b$,

$$c_1 u_1 v_1' + c_2 u_2 v_2' = -\frac{1}{2}x\cdot 2x + 1\cdot(x^2+1)\cdot 1 = 1 = a,$$

and therefore in fact $(-\frac{1}{2}\log(x^2+1) + \log x)' = 1/(x^3+x)$, as in Example 22.6. $\diamond$

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Although Theorem 22.8 keeps the degree of the algebraic extension as small as possible, it does not avoid gcd computations in such an algebraic extension. The following observation, however, will lead to a purely rational algorithm.

THEOREM 22.9 *Lazard-Rioboo-Trager formula.* *Let $a, b, c_1, \ldots, c_l, v_1, \ldots, v_l$ be as in Theorem 22.8, $i \in \{1, \ldots, l\}$, and let $e \geq 1$ denote the multiplicity of c_i as a root of the resultant $r = \operatorname{res}_x(b, a - yb')$.*

(i) $\deg v_i = e$.

(ii) Let $w(x, y) \in F(y)[x]$ denote the remainder of degree e in the monic Euclidean Algorithm for b and $a - yb'$ in $F(y)[x]$. Then $v_i = w(x, c_i)$.

PROOF. We let $\lambda_1, \ldots, \lambda_n$ be the roots of b in its splitting field. Then

$$r = \operatorname{lc}_x(b)^{\deg_x(a-yb')} \prod_{1 \leq k \leq n} (a(\lambda_k) - yb'(\lambda_k)),$$

by Exercise 6.12, and

$$e = \#\{k \in \mathbb{N} : 1 \leq k \leq n \text{ and } a(\lambda_k) - c_i b'(\lambda_k) = 0\}.$$

On the other hand,

$$v_i = \gcd(b, a - c_i b') = \gcd\left(\prod_{1 \leq k \leq n} (x - \lambda_k), a - c_i b'\right) = \prod_{\substack{1 \leq k \leq n \\ a(\lambda_k) - c_i b'(\lambda_k) = 0}} (x - \lambda_k),$$

since the λ_k are distinct. This proves (i).

For (ii), we let $\sigma \in F[y]$ denote the eth subresultant (with respect to x) of b and $a - yb'$. If $e < n$, then (i) and Theorem 6.55 with $R = F[y]$ and $p = y - c_i$ imply that $\sigma(c_i) \neq 0$ and $w(x, c_i)$ is well-defined and equal to v_i. If $e = n$, then $w(x, c_i) = w(x, y) = b(x)/\operatorname{lc}(b)$, and (i) implies that $v_i = b/\operatorname{lc}(b)$ as well. □

Thus it is sufficient to separate the roots of the resultant r by their multiplicities, that is, to compute the squarefree decomposition of r; the complete irreducible factorization is not necessary. This leads to the following integration algorithm for rational functions.

ALGORITHM 22.10 Symbolic integration of rational functions.
Input: $f, g \in F[x]$ coprime, where F is a field of characteristic zero and $g \neq 0$ is monic.
Output: $\int \frac{f}{g}$.

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1. $h \longleftarrow f \text{ rem } g, \quad \sum_i q_i x^i \longleftarrow f \text{ quo } g, \quad U \longleftarrow \sum_i \frac{q_i}{i+1} x^{i+1}$

2. **call** Yun's algorithm 14.21 to compute the squarefree decomposition $g = \prod_{1 \le e \le m} g_i^i$ of g

3. compute the partial fraction decomposition $\frac{h}{g} = \sum_{1 \le i \le m} \sum_{1 \le j \le i} \frac{h_{ij}}{g_i^j}$, with $h_{ij} \in F[x]$ such that $\deg h_{ij} < \deg g_i$ for $1 \le j \le i \le m$

4. { Hermite reduction }
   $V \longleftarrow 0$
   for $i = 2, \ldots, m$ **do**
      for $j = i, i-1, \ldots, 2$ **do**
         compute $s, t \in F[x]$ such that $sg_i + tg_i' = h_{ij}$ and $\deg s, \deg t < \deg g_i$ using Theorem 4.10
         $V \longleftarrow V - \frac{t}{(j-1)g_i^{j-1}}, \quad h_{i,j-1} \longleftarrow h_{i,j-1} + s + \frac{t'}{j-1}$

5. $W \longleftarrow 0$
   for $i = 1, \ldots, m$ **do**

6. { Lazard-Rioboo-Trager method }
   $a \longleftarrow h_{i1}, \quad b \longleftarrow g_i, \quad r \longleftarrow \text{res}_x(b, a - yb')$
   call Yun's algorithm 14.21 to compute the squarefree decomposition $r = \text{lc}(r) \prod_{1 \le e \le d} r_e^e$ of r

7. **for** $e = 1, \ldots, d$ such that $\deg r_e > 0$ **do**
      compute the remainder $w_e(x, y) \in F(y)[x]$ of degree e in the monic Euclidean Algorithm of b and $a - yb'$ in $F(y)[x]$
      $W \longleftarrow W + \sum_{r_e(\gamma)=0} \gamma \log w_e(x, \gamma)$

8. **return** $U + V + W$

Updating V in step 4 may be done by bringing $V - t/((j-1)g_i^{j-1})$ to a common denominator. Usually, however, it is more desirable to concatenate the summands symbolically, without actually computing the sum. This also applies to the updating of W in step 7, and then the integral is returned as a formal sum of several "small" rational functions and logarithms in step 8. Many computer algebra systems handle it that way, and the user must explicitly direct the system to bring the integral to a common denominator after calling the symbolic integration engine. A similar remark applies to the sum over all zeroes γ of r_e in step 7. This can be

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left as is, without further evaluation (for example, by using the `RootOf` construct in MAPLE).

EXAMPLE 22.6 (continued). Let $f = 1$ and $g = x^2 - 2$, as in Example 22.6 (ii). Then $g = g_1$ is squarefree, and nothing happens in steps 1 through 4, so that $U = V = 0$. In step 6, we have $m = 1$, $a = f$, $b = g$, and

$$r = \operatorname{res}_x(x^2 - 2, 1 - y \cdot 2x) = \det \begin{pmatrix} 1 & -2y & 0 \\ 0 & 1 & -2y \\ -2 & 0 & 1 \end{pmatrix} = -8y^2 + 1.$$

This polynomial is irreducible in $\mathbb{Q}[x]$, so that $d = 1$ and $r_1 = r/\operatorname{lc}(r) = y^2 - 1/8$. The monic remainder of degree 1 in x in the Euclidean Algorithm over $\mathbb{Q}(y)$ for b and $a - yb' = -2yx + 1$ is simply the latter polynomial divided by its leading coefficient, namely $w_1 = x - 1/2y$. Thus

$$W = \sum_{r_1(\gamma)=0} \gamma \log(x - 1/2\gamma),$$

and if we plug in the two zeroes $\pm 1/2\sqrt{2} \in \mathbb{R}$ of r_1 for γ, we arrive at the same result as in Example 22.6. $\diamond$

THEOREM 22.11.
Algorithm 22.10 works correctly as specified. If f and g are of degree at most n, its running time is $O(n\mathsf{M}(n)\log n)$ or $O^{\sim}(n^2)$ operations in F.

PROOF. Correctness follows from the discussion preceding Theorem 22.7 and from Theorems 22.8 and 22.9.

By Theorem 22.7, steps 1 through 4 take $O(\mathsf{M}(n)\log n)$ operations. Let $i \in \{1, \ldots, m\}$ and $d_i = \deg g_i$. Exercise 6.12 implies that $\deg r = d_i$, and the cost for computing r is $O(d_i\mathsf{M}(d_i)\log d_i)$, by Corollary 11.21. Within the same time bound, we can in fact compute all the w_e with $\deg r_e > 0$ in step 7, by Exercise 11.9, since

$$\sum_{\substack{1 \le e \le d \\ \deg r_e > 0}} e \le \sum_{1 \le e \le d} e \deg r_e = \deg r = d_i.$$

Now $\sum_{1 \le i \le m} d_i \le n$, and the overall cost for the loop 5 is $O(n\mathsf{M}(n)\log n)$ operations in F. This dominates the cost for the other steps, and the claim follows. $\square$

22.4. Hyperexponential integration: Almkvist & Zeilberger's algorithm

The preceding sections provide a complete algorithm for symbolic integration of rational functions. In this section, we present an algorithm for the larger class

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of hyperexponential functions. In contrast to the method for rational functions, however, this algorithm does not always return a symbolic integral, but only if the integral is again hyperexponential.

DEFINITION 22.12. *Let (L,D) be a differential field, $K \subseteq L$ a subfield, and $f \in L$ with* ***logarithmic derivative*** $D(f)/f \in K$. *Then f is* ***hyperexponential over K***.

EXAMPLE 22.13. (i) Every rational function in $\mathbb{Q}(x)$ is hyperexponential over $\mathbb{Q}(x)$ with respect to the usual differential operator $Dx = 1$.

(ii) Let $L = \mathbb{Q}(x, \exp(x^2))$ with $Dx = 1$ and $D(\exp(x^2)) = 2x \cdot \exp(x^2)$. Then $f = \exp(x^2)$ is hyperexponential over $\mathbb{Q}(x)$.

(iii) The square root function $f = x^{1/2} \in L = \mathbb{Q}(x, x^{1/2})$ is hyperexponential over $\mathbb{Q}(x)$, with logarithmic derivative $D(x^{1/2})/x^{1/2} = 1/2x$. $\diamond$

In the remainder of this chapter, "hyperexponential" always means "hyperexponential over $K = F(x)$", where F is a field of characteristic zero, and we assume that $D = '$ is the usual derivative on $F(x)$. The **hyperexponential integration problem** is to determine whether a nonzero hyperexponential element g in an extension field of $F(x)$ has a hyperexponential integral, and if so, to compute one.

We assume that g' is nonzero and let $\sigma = g'/g \in F(x)$ be the logarithmic derivative of g. If f is another hyperexponential element with logarithmic derivative $\rho = f'/f \in F(x)$ such that $g = f' = \rho f$, then $f = g/\rho$ is a rational multiple of g. (We have $\rho \neq 0$ since $g' \neq 0$ implies $f' \neq 0$.) Thus, if g has a hyperexponential integral, then this already belongs to the same differential field as g.

Let $\tau \in F(x)$ and $f = \tau g$. Then

$$f' = (\tau g)' = \tau' g + \tau g' = (\tau' + \sigma\tau) g,$$

and this equals g if and only if τ satisfies the first order differential equation

$$\tau' + \sigma\tau = 1. \tag{4}$$

We note that only rational functions occur in (4). Thus we have eliminated all hyperexponential elements and reduced the original task to a purely rational problem. In what follows, we discuss how to solve the differential equation (4).

We write $\sigma = a/b$ and $\tau = u/v$, with nonzero polynomials $a, b, u, v \in F[x]$ such that b and v are monic and $\gcd(a,b) = \gcd(u,v) = 1$. Then (4) becomes

$$\tau' + \sigma\tau = \frac{u'v - uv'}{v^2} + \frac{au}{bv} = 1.$$

After multiplying by bv^2, we obtain the equivalent polynomial differential equation

$$bu'v - buv' + auv = bv^2 \tag{5}$$

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Conversely, any solution $u, v \in F[x]$ of (5) for given $a, b \in F[x]$ yields a solution of our hyperexponential integration problem by setting $f = gu/v$.

The algorithm proceeds in two phases. In the first phase, we find a multiple V of any possible denominator v. Once we know v (or a multiple of it), then (5) is just a system of linear equations in the coefficients of u. It is easily solved if we can bound the degree of u; such a bound is calculated in Lemma 22.18 below.

For the first phase, we want to determine a suitable multiple of v. Let $v_0, v_1 \in F[x]$ be such that $v = v_0 \cdot \gcd(v, v')$ and $v' = v_1 \cdot \gcd(v, v')$. Dividing (5) by $\gcd(v, v')$, we find

$$bu'v_0 - buv_1 + auv_0 = bvv_0. \tag{6}$$

We see that v_0 divides buv_1, and since $\gcd(u, v) = 1 = \gcd(v_0, v_1)$, it divides b. Thus we can divide (6) by v_0 and obtain

$$bu' + \left(a - \frac{b}{v_0}v_1\right)u = bv.$$

Again, since v_0 divides b and is coprime to u, we conclude that v_0 divides $a - (b/v_0)v_1$.

Now we let $(h_1, \ldots, h_m)$ be the squarefree decomposition of v, as defined in Section 14.6, with $m \in \mathbb{N}_{\geq 1}$ and monic squarefree and pairwise coprime polynomials $h_1, \ldots, h_m \in F[x]$ such that $v = h_1 h_2^2 \cdots h_m^m$ and $h_m \neq 1$. Then $v_0 = h_1 \cdots h_m$ and $v_1 = \sum_{1 \leq i \leq m} i h_i' v_0 / h_i$, by Exercise 14.26. Letting $1 \leq i \leq m$ and computing modulo h_i, we find

$$\frac{b}{v_0}v_1 \equiv \frac{b}{v_0} \cdot i h_i' \frac{v_0}{h_i} = i\frac{b}{h_i}h_i' \equiv i\left(\frac{b}{h_i}h_i' + \left(\frac{b}{h_i}\right)' h_i\right) = ib' \bmod h_i.$$

Now h_i divides b, by the above, and both left hand summands in

$$\left(a - \frac{b}{v_0}v_1\right) + \left(\frac{b}{v_0}v_1 - ib'\right) = a - ib',$$

so that h_i divides $\gcd(b, a - ib')$, for $1 \leq i \leq m$. This leads to the following algorithm for computing a multiple of v.

ALGORITHM 22.14 Multiple of integration denominator.
Input: Relative prime polynomials $a, b \in F[x]$ with $b \neq 0$ monic.
Output: A monic polynomial $V \in F[x]$ such that for any coprime $u, v \in F[x]$, equation (5) implies that v divides V.

1. $R \longleftarrow \operatorname{res}_x(b, a - yb')$, $\quad d \longleftarrow \max\{i \in \mathbb{N} : i = 0 \text{ or } R(i) = 0\}$
   if $d = 0$ **then return** 1
2. **for** $i = 1, \ldots, d$ **do** $H_i \longleftarrow \gcd(b, a - ib')$
3. **return** $H_1 H_2^2 \cdots H_d^d$

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We note that the resultant in step 1 is the same as in algorithm 22.10. However, here we do not need its complete factorization into irreducible polynomials, but only its positive integral roots.

THEOREM 22.15.
Algorithm 22.14 works correctly as specified. More precisely, if u, v are coprime polynomials in $F[x]$ solving (5), $v \neq 0$ is monic, and $(h_1, \ldots, h_m)$ is the squarefree decomposition of v, then $m \leq d$ and h_i divides H_i for $1 \leq i \leq m$.

PROOF. We may assume that $\deg v \geq 1$. We have $\deg h_m > 0$, by the definition of the squarefree decomposition. By the discussion preceding the theorem, h_i divides $\gcd(b, a - ib')$ for all i. In particular, this implies that $\gcd(b, a - mb')$ is nonconstant, and hence $R(m) = \operatorname{res}_x(b, a - mb') = 0$. Thus $m \leq d$ and $h_i \mid H_i$ for $1 \leq i \leq m$. Finally, $v = h_1 h_2^2 \cdots h_m^m$ divides $H_1 H_2^2 \cdots H_m^m H_{m+1}^{m+1} \cdots H_d^d = V$. □

We note that it is sufficient to perform the loop in step 2 only for those i that are roots of the resultant R. Moreover, it is somewhat more efficient to remove H_i from b and $a - ib'$ in step 2; see Exercise 22.11.

EXAMPLE 22.16. Let $g = (x^3 + x^2)\exp(x) \in \mathbb{Q}(x, \exp(x)) = L$. The logarithmic derivative of g is

$$\frac{g'}{g} = \frac{(3x^2 + 2x)\exp(x) + (x^3 + x^2)\exp(x)}{(x^3 + x^2)\exp(x)} = \frac{x^2 + 4x + 2}{x^2 + x},$$

so that g is hyperexponential over $K = \mathbb{Q}(x)$, and we have $a = x^2 + 4x + 2$ and $b = x^2 + x$. Algorithm 22.14 computes

$$R = \operatorname{res}_x(x^2 + x, x^2 + 4x + 2 - y(2x + 1)) = \det \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 4 - 2y & 1 \\ 0 & 1 & 2 - y & 4 - 2y \\ 0 & 0 & 0 & 2 - y \end{pmatrix}$$
$$= -y^2 + 3y - 2 = -(y - 1)(y - 2),$$

so that $d = 2$. In step 2 of Algorithm 22.14, we compute

$$H_1 = \gcd(b, a - b') = \gcd(x^2 + x, x^2 + 2x + 1) = x + 1,$$
$$H_2 = \gcd(b, a - 2b') = \gcd(x^2 + x, x^2) = x,$$

and finally $V = H_1 H_2^2 = x^3 + x^2$ in step 3. ◇

The following example shows that d may be exponentially large in the input size.

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EXAMPLE 22.17. We let $g = x^n \exp(x) \in \mathbb{Q}(x, \exp(x))$, with a parameter $n \in \mathbb{N}$. Then g is hyperexponential over $\mathbb{Q}(x)$, with logarithmic derivative

$$\frac{g'}{g} = \frac{x+n}{x},$$

and the input to Algorithm 22.14 is $a = x+n$ and $b = x$. Now $\gcd(b, a - ib') = \gcd(x, x+n-i)$ is nonconstant precisely when $i = n$, and hence $d = n$ in step 1, $H_i = 1$ for $1 \le i < n$, and $H_n = x$, so that $V = x^n$. The degree of V is exponential in the size of a and b, which is about $\log_{2^{64}} n$ words. However, it is polynomial in the "degree" of g with respect to x. $\diamond$

Now that we have found a monic multiple V of the denominator v, it remains to find a suitable numerator polynomial $U \in F[x]$ such that

$$bVU' - (bV' - aV)U = bV^2. \tag{7}$$

Then $\tau = U/V \in F(x)$ is a solution of (4), and $f = gU/V$ is a hyperexponential integral of g. Conversely, if $\tau = u/v$ solves (4), with coprime polynomials u, v and $v \neq 0$ monic, then $U = uV/v$ is a solution of (7).

We divide both sides of (7) by $h = \gcd(bV, bV' - aV) \in F[x]$ and obtain the equivalent equation

$$rU' - sU = t, \tag{8}$$

where $r = bV/h$ is nonzero and monic, $s = (bV' - aV)/h$, and $t = rV$ is nonzero. (We can actually achieve $t = V$; see Exercise 22.12.) Suppose that we know the degree of U, or at least an upper bound. Then we can plug a generic polynomial U with unknown coefficients into (8) and solve the resulting system of linear equations for the coefficients of U. The following lemma provides such a degree bound.

LEMMA 22.18. *Let $r, s, t, U \in F[x]$ satisfy (8), with r, U nonzero and r monic, let $m = \max\{\deg r - 1, \deg s\}$, and let $\delta \in F$ be the coefficient of x^m in s. (As usual, $\delta = 0$ if $m < 0$ or $\deg s < m$.) Moreover, let*

$$e = \begin{cases} \deg t - m & \text{if } \deg r - 1 < \deg s \text{ or } \delta \notin \mathbb{N} \setminus \{0, 1, \ldots, \deg t - m\}, \\ \delta & \text{otherwise.} \end{cases}$$

Then the following hold.

(i) Either $\deg U = \deg t - m$, or else $\deg U = \delta > \deg t - m$ and $\deg r - 1 \ge \deg s$. In particular, $\deg U \le e$.

(ii) If $\deg r - 1 \ge \deg s$, then $\deg t - m \neq \delta$.

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(iii) *If* $t = 0$, *then* $\deg r - 1 \geq \deg s$ *and* $\deg U = \delta \in \mathbb{N}$.

(iv) *If* $\deg r - 1 < \deg s$ *or* $\delta \notin \mathbb{N}$, *then exactly one* $U \in F[x]$ *satisfies equation (8).*

We recall that the zero polynomial has degree $-\infty$; thus $\deg t - m$ is to be interpreted as $-\infty$ if $t = 0$. We have $\mathbb{N} \subseteq \mathbb{Q} \subseteq F$, so that any integer is also an element of F.

PROOF. We compare the degrees and the top coefficients in (8). Firstly, we have

$$\begin{aligned} \deg t &\leq \max\{\deg(rU'), \deg(sU)\} \\ &\leq \max\{\deg r + \deg U - 1, \deg s + \deg U\} = m + \deg U. \end{aligned}$$

Let $\gamma \in F$ denote the coefficient of x^{m+1} in r. Then the coefficient of $x^{m+\deg U}$ in rU' is $\gamma \operatorname{lc}(U) \deg U$, and the coefficient of $x^{m+\deg U}$ in sU is $\delta \operatorname{lc}(U)$. Thus the coefficient of $x^{m+\deg U}$ in t is $(\gamma \deg U - \delta)\operatorname{lc}(U)$, and $\deg t < m + \deg U$ if and only if this coefficient vanishes.

If $\deg r - 1 < \deg s$, then $\gamma = 0$ and $\delta = \operatorname{lc}(s) \neq 0$, and hence $\deg U = \deg t - m$. Otherwise, we have $\gamma = \operatorname{lc}(r) = 1$. We conclude that $\deg U \geq \deg t - m$, with strict inequality if and only if $\deg r - 1 \geq \deg s$ and $\deg U = \delta$. This proves (i), (ii), and (iii).

To show (iv), we assume that $U^* \in F[x]$ is another solution of (8). Then the difference $U - U^*$ satisfies the homogeneous equation $r(U - U^*)' - s(U - U^*) = 0$, and the claim follows from (iii). □

Thus we can (almost) determine $\deg U$ from the known polynomials r, s and $t = rV$, namely $\deg U = e = \deg t - m$ if $\deg r - 1 < \deg s$, and $\deg U \leq e = \max(\{\deg t - m, \delta\} \cap \mathbb{Z})$ if $\deg r - 1 \geq \deg s$. If the bound e is nonnegative and if $\deg r - 1 < \deg s$ or $\deg t - m \neq \delta$, we set up the system of linear equations equivalent to (8) and solve it for the unknown coefficients of U. If the system has a solution, then $\tau = U/V$ satisfies (4), and gU/V is a hyperexponential integral of g. Otherwise, or if $e < 0$, or if $\deg r - 1 \geq \deg s$ and $\deg t - m = \delta$, then we know that (4) has no rational solution $\tau \in F(x)$, and hence g has no hyperexponential integral.

ALGORITHM 22.19 Hyperexponential integration.

Input: Nonzero coprime polynomials $a, b \in F[x]$, with $b \neq 0$ monic, where F is a field of characteristic zero.

Output: Coprime polynomials $u, v \in F[x]$, with v monic, satisfying (5), if such polynomials exists, and otherwise "unsolvable".

1. **call** Algorithm 22.14 with input a, b, yielding a monic multiple $V \in F[x]$ of any possible denominator

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2. $h \longleftarrow \gcd(bV, bV' - aV)$, $\quad r \longleftarrow \dfrac{bV}{h}$, $\quad s \longleftarrow \dfrac{bV' - aV}{h}$, $\quad t \longleftarrow r \cdot V$

3. $m \longleftarrow \max\{\deg r - 1, \deg s\}$
   let δ be the coefficient of x^m in s
   if $\deg r - 1 < \deg s$ or $\delta \notin \mathbb{N}$ **then** $e \longleftarrow \deg t - m$
   else if $\deg t - m = \delta$ **then return** "unsolvable"
   else $e \longleftarrow \max\{\deg t - m, \delta\}$
   if $e < 0$ **then return** "unsolvable"

4. solve the linear system corresponding to (8) for the unknown coefficients $U_0, \ldots, U_e \in F$ of U of degree at most e
   if the system is unsolvable **then return** "unsolvable"
   else $U \longleftarrow U_e x^e + \cdots + U_1 x + U_0$

5. **return** $\dfrac{U}{\gcd(U,V)}$ and $\dfrac{V}{\gcd(U,V)}$

The solution space of the system of linear equations in step 4 is either empty, or has precisely one element, or turns out to be one-dimensional (Exercise 22.13). In the latter case, we preferably take a solution leading to a numerator U of smallest degree. The coefficient matrix is triangular, so that the system is particularly easy to solve without Gaussian elimination, simply by back substitution, taking $O(e^2)$ operations in F. At most one diagonal element is zero, and this occurs only if $\deg r - 1 \geq \deg s$ and $\delta \in \mathbb{N}$, by Lemma 22.18 (iii).

Exercise 22.12 shows that we may even obtain $\gcd(r,t) = 1$ in step 2, by appropriately dividing out common factors, and perform steps 3 through 5 for a divisor of U. This may further reduce the size of the system in step 4.

We now give some examples.

EXAMPLE 22.17 (continued). We recall that $a = x + n$, $b = x$, and $V = x^n$, where $n \in \mathbb{N}$ is a parameter. Now $bV = x^{n+1}$, $bV' - aV = -x^{n+1}$, and hence $h = x^{n+1}$, $r = 1$, $s = -1$, and $t = x^n$ in step 2 of Algorithm 22.19. Thus equation (8) reads

$$U' + U = x^n. \tag{9}$$

We have $m = \max\{\deg r - 1, \deg s\} = 0 > \deg r - 1$, so that we are in the first case of Lemma 22.18, and the degree bound in step 3 is $e = \deg t - m = n$. In step 4, we let $U = U_n x^n + \cdots + U_1 x + U_0$, with undetermined coefficients $U_n, \ldots, U_0$. Plugging this into (9), we obtain the system of linear equations

$$U_n = 1, \quad nU_n + U_{n-1} = 0, \quad \ldots, \quad 2U_2 + U_1 = 0, \quad U_1 + U_0 = 0.$$

This system has a unique solution, and it satisfies $U_0 \neq 0$. Finally, the algorithm returns $u = U$ and $v = V$ in step 5.

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For example, for $n = 2$ the solution is $U_2 = 1$, $U_1 = -2$, and $U_0 = 2$, and hence $u = U = x^2 - 2x + 2$. We check that gu/v is indeed an integral of $g = x^2 \exp(x)$:

$$\left(g\frac{u}{v}\right)' = \Big((x^2 - 2x + 2)\exp(x)\Big)' = (2x-2)\exp(x) + (x^2 - 2x + 2)\exp(x) = g. \diamond$$

EXAMPLE 22.20. Let $g = 1/x \in F(x)$. Then $\sigma = g'/g = -1/x$, and $a = -1$ and $b = x$. We compute $R = \operatorname{res}_x(x, -1 - y) = -(y+1)$ in step 1 of Algorithm 22.14 and hence $d = 0$ and $V = 1$. Plugging this into (8), we obtain

$$xU' - U = x.$$

We have $r = t = x$, $s = 1$, $m = \deg s = \deg r - 1 = 0$, and $\deg t - m = 1 = \delta$. By Lemma 22.18 (ii), the above equation has no solution $U \in F[x]$. We conclude that $\int(1/x)$ is not hyperexponential. This can also be checked directly: the logarithmic derivative of $\log x = \int(1/x)$ is $1/x \log x$, and this is not a rational function. $\diamond$

EXAMPLE 22.21. This example illustrates that $\deg U = \delta$ may actually occur. Let $g = (x^2+1)^{-3/2} \in \mathbb{Q}(x, \sqrt{x^2+1})$. Then g is hyperexponential over $\mathbb{Q}(x)$ with logarithmic derivative

$$\frac{g'}{g} = \frac{-3x}{x^2+1},$$

so that $a = -3x$ and $b = x^2 + 1$. In step 1 of Algorithm 22.14, we compute

$$R = \operatorname{res}_x(x^2+1, -3x - y \cdot 2x) = \det \begin{pmatrix} 1 & -3-2y & 0 \\ 0 & 0 & -3-2y \\ 1 & 0 & 0 \end{pmatrix} = (2y+3)^2,$$

so that $d = 0$ and $V = 1$. Now equation (8) reads

$$(x^2+1)U' - 3xU = x^2 + 1. \tag{10}$$

Thus $r = t = b = x^2 + 1$, $s = -a = 3x$, $m = \max\{\deg r - 1, \deg s\} = 1$, and we are in the second case of Lemma 22.18. Now $1 = \deg t - m \neq \delta = 3$, and the lemma implies that $\deg U = \delta = 3$. We let $U = U_3x^3 + U_2x^2 + U_1x + U_0$, with undetermined coefficients $U_3, \ldots, U_0$. Plugging this into (10), we see that the coefficients of x^4 cancel, and comparing coefficients of $x^3, x^2, x, 1$ leads to the linear equations

$$-U_2 = 0, \quad 3U_3 - 2U_1 = 1, \quad 2U_2 - 3U_0 = 0, \quad U_1 = 1.$$

The unique solution of these equations is $U_3 = U_1 = 1$ and $U_2 = U_0 = 0$. Thus $U = x^3 + x$ in step 4, and the algorithm returns $u = U$ and $v = 1$ in step 5. We check that gu/v is an integral of g:

$$\begin{aligned}\left(g\frac{u}{v}\right)' &= \Big(x \cdot (x^2+1)^{-1/2}\Big)' = (x^2+1)^{-1/2} - \frac{1}{2}x \cdot (x^2+1)^{-3/2} \cdot 2x \\ &= (x^2+1)^{-3/2}(x^2 + 1 - x^2) = g. \diamond\end{aligned}$$

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Notes. Historically, the foundations of symbolic integration were laid by Joseph Liouville (1833a, 1833b, 1835). Ritt (1948) invented the notion of a differential algebra, which is the appropriate framework for the integration problem. A more general method was presented by Risch (1969, 1970), and variants of his algorithm are implemented today in almost any computer algebra system. They employ suitably modified versions of Hermite's and Rothstein's and Trager's methods.

Richardson (1968) and Caviness (1970) showed that a sufficiently general version of the integration problem is unsolvable. Already when we just consider real functions built up from the constant 1, a single variable, the four arithmetic operations, and the sine function, then determining whether the definite integral (from $-\infty$ to ∞) of such an expression exists is undecidable, and similarly for the existence of indefinite integrals as we considered in this chapter (Matiyasevich 1993, §9.4).

In spite of this fundamental limitation, symbolic integration and, more generally, the symbolic solution of ordinary differential equations, is a highly active area of research. Among its goals are algorithms for a wider range of problems, and better algorithms for special types of problems. Bronstein (1997) gives a nice overview.

22.1. A project for OCR-reading integral tables is described in Berman & Fateman (1994).

22.2. Most undergraduate calculus textbooks contain an integration algorithm for rational functions by factoring the denominator into linear polynomials over the complex numbers (or at most quadratic polynomials over the real numbers) and performing a complete partial fraction decomposition. For rational functions with only simple poles, this algorithm first appears in Johann Bernoulli (1703). For symbolic computation, this approach is inefficient since it involves polynomial factorization and computation with algebraic numbers, and most computer algebra systems implement the algorithms described in this chapter.

In fact, on the pages just preceding Bernoulli's article, Leibniz (1703) goes a step further. He also computes the integral of a rational function by partial fraction decomposition if the denominator is a product of distinct linear factors. But then he also gives the decomposition in several cases where the denominator is not squarefree, for example,

$$\frac{1}{h^4 l} = \frac{1}{\omega h^4} - \frac{1}{\omega^2 h^3} + \frac{1}{\omega^3 h^2} - \frac{1}{\omega^4 h} + \frac{1}{\omega^4 l},$$

where $h = x + a$, $l = x + b$, and $\omega = b - a$, similarly for $1/h^4 l^3$, and a general formula for $1/h^t l^s$. For those terms, *interveniunt etiam Hyperboloidum quadraturæ, quales sunt, quarum ordinatæ sunt* $\frac{1}{xx}, \frac{1}{x^3}, \frac{1}{x^4}$, *&c*,[1] which are calculated by the usual rules. Here we have the essential ingredients of Hermite's method, 170 years before Hermite!

Ostrogradsky (1845) shows that for coprime $f, g \in F[x]$ with $\deg f < \deg g$, there exist unique polynomials $a, c \in F[x]$ such that (1) holds, $\deg a < \deg b$, and $\deg c < \deg d$, where b is the squarefree part of g and $d = g/b$, and presents an algorithm for computing a and c. The algorithm described in Section 22.2 is from Hermite (1872). Theorem 22.7 appears in Yun (1977a). Gerhard (2001a) gives fast modular algorithms for Hermite reduction.

22.3. Theorem 22.8 is from Rothstein (1976, 1977) and Trager (1976). Theorem 22.9 is from Lazard & Rioboo (1990). Independently, Trager implemented the resulting algorithm in SCRATCHPAD, but did not publish it. Mulders (1997) describes an error in software implementations of the Lazard-Rioboo-Trager method.

[1] furthermore enter the integrals of the hyperbolas, whose function values are x^{-2}, x^{-3}, x^{-4}, etc.

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Gerhard (2001a) presents a modular variant of Algorithm 22.10. If $f, g \in \mathbb{Z}[x]$ and their coefficients are absolutely bounded by $A \in \mathbb{N}$, then this algorithm takes $O^{\sim}(n^5 + n^4 \log A)$ word operations.

22.4. Algorithm 22.19 is from Almkvist & Zeilberger (1990). It is the analog of Gosper's (1978) algorithm for hypergeometric summation, which we discuss in Section 23.4. The method of Almkvist & Zeilberger for determining a multiple of the denominator is described in Exercise 22.11.

Equation (4) is a special case of the *Risch differential equation*: given elements σ, ρ in a differential field K, determine $\tau \in K$ satisfying $D\tau + \sigma\tau = \rho$. Equation (4) corresponds to the case $K = F(x)$, $\rho = 1$, and $D = {}'$. The more general equation plays a significant role in Risch's (1969, 1970) algorithm. Other algorithms for solving the Risch differential equation are given by Rothstein (1976, 1977), Kaltofen (1984), Davenport (1986), and Bronstein (1990, 1991, 1997). Algorithms computing rational solutions of linear differential equations of higher order are given by Abramov (1989a, 1989b), Abramov & Kvashenko (1991), Bronstein (1992), Abramov, Bronstein & Petkovšek (1995), and Bronstein & Fredet (1999). The algorithms of Singer (1991), Petkovšek & Salvy (1993), and Pflügel (1997), for example, find more general solutions of higher order equations.

The idea of finding polynomial solutions of linear differential equations by first determining a degree bound and then solving a system of linear equations is known as the *method of undetermined coefficients* and goes back to Newton (1691/92). Gerhard (2001b), Chapter 9, presents several asymptotically fast methods for first order equations.

In practice, one would compute all integral roots of the resultant R in step 1 of Algorithm 22.14, using Corollary 11.21 and Theorem 15.21, and then iterate the loop in step 2 only for those i that are roots of the resultant. Gerhard (2001b), Section 8.1 and Chapter 10, gives a cost analysis of modular variants of Algorithms 22.14 and 22.19, respectively, in terms of word operations.

Whether the output of Algorithm 22.14 is exponential in the input size depends on what we regard as input. For example, if we consider hyperexponential functions of the form $g = r_1 \exp(r_2)$ with $r_1, r_2 \in \mathbb{Q}(x)$, then the result returned by Algorithm 22.14 is polynomial in the degrees of the numerators and denominators of r_1 and r_2 (Theorem 10.16 in Gerhard 2001b). If we think of this algorithm as the first step of an integration algorithm, then it is natural to regard g—and not the logarithmic derivative g'/g—as input. If we represent g by storing the rational functions r_1 and r_2 in a dense format as quotients of polynomials, then the output of Algorithm 22.14 is polynomial in the input size. However, if we represent g in a sparse format, e.g., by an "expression tree", or if we consider the logarithmic derivative g'/g as input, then Example 22.17 shows that the output may be exponential in this input size. On the other hand, Algorithm 22.14 may also be regarded as the first step of a method for solving the linear differential equation (4), and then it is natural to consider $\sigma = g'/g$ as the input. In fact, for a higher order linear differential equation with rational function coefficients, the degrees of the rational solutions are in general exponential in the size of the coefficients of the differential equation. Similar remarks also apply to the degree of the numerator in Algorithm 22.19.

Exercises.

22.1 Let (R, D) be a differential algebra. Show that R_0 is in fact a subring of R, and a subfield if R is a field.

https://doi.org/10.1017/CBO9781139856065.028 Published online by Cambridge University Press

22.2 Prove Lemma 22.2.

22.3 Show that $\mathbb{Q}$ has only the trivial derivative.

22.4 Let $'$ denote the usual derivative on $\mathbb{Q}(x)$. Prove *without* using arguments from calculus that $f' = 0 \Longrightarrow f \in \mathbb{Q}$ holds for all $f \in \mathbb{Q}(x)$. Hint: Prove the claim first for polynomials $f \in \mathbb{Q}[x]$.

22.5* Let F be a field of characteristic zero, and $a, b, c, d \in F[x] \setminus \{0\}$ such that $(c/d)' = a/b$.

(i) Prove that $\deg a - \deg b \leq \deg c - \deg d - 1$, with equality if and only if $\deg c \neq \deg d$. Give an example where equality does not hold. Conclude that $\deg a - \deg b = -1$ is impossible.

(ii) Let $p \in F[x]$ be irreducible and $v_p(a) = e \in \mathbb{N}$ if $p^e \mid a$ and $p^{e+1} \nmid a$ (this is the negative logarithm of the p-adic value of a, as in Example 9.31 (iii)), and similarly $v_p(b)$, $v_p(c)$, $v_p(d)$. Prove that $v_p(a) - v_p(b) \geq v_p(c) - v_p(d) - 1$, with equality if and only if $v_p(c) \neq v_p(d)$. Give an example where equality does not hold. Conclude that $v_p(a) - v_p(b) = -1$ is impossible, and that $v_p(b) \geq 2$ for every irreducible divisor p of b if $\gcd(a, b) = 1$. In particular, b is not squarefree if it is nonconstant and coprime to a.

22.6* You are to show that e^{x^2} has no integral of the form $g \cdot e^{x^2}$ with a rational function $g \in \mathbb{R}(x)$.

(i) Suppose to the contrary that such a g exists. Derive a first order differential equation $g' + sg = t$ for g, with polynomials $s, t \in \mathbb{R}[x]$.

(ii) Now assume that $g = u/v$ with coprime $u, v \in \mathbb{R}[x]$ and v monic. Prove that $v = 1$.

(iii) Compare degrees on both sides of the equation $u' + su = t$, and conclude that no polynomial $u \in \mathbb{R}[x]$ satisfies it.

22.7* Let F be a field of characteristic zero and $f, g \in F[x]$ nonzero polynomials with g monic.

(i) Prove that the decomposition (1) is unique (up to adding a constant to c/d) when we stipulate that b is the (monic) squarefree part of g, $d = f/b$, and $\deg a < \deg b$. Hint: Use Exercise 22.5 (ii).

(ii) Show that a decomposition (1) always exists uniquely if $g = bd$ for monic polynomials $b, d \in F[x]$ such that $b \mid d$ and every irreducible factor of d divides b, and $\deg a < \deg b$. Give an example where such a decomposition does not exist if we drop the requirement that every irreducible factor of d divides b.

22.8* This exercise discusses a variant of Hermite reduction, due to Mack (1975) (see also Bronstein 1997, §2.2). Let F be a field of characteristic zero and $g, h \in F[x]$ nonzero such that $n = \deg g > \deg h$ and g is monic.

(i) Let $g = g_1 g_2^2 \cdots g_m^m$ be the squarefree decomposition of g, with all $g_i \in F[x]$ monic, pairwise coprime, and $g_m \neq 1$. Moreover, let $g^* = g/g_1$, $b = g_2 \cdots g_m$ be the squarefree part of g^*, and $d = g_2 g_3^2 \cdots g_m^{m-1} = g^*/b$. Now $\gcd(g_1, g^*) = 1$, and we have the partial fraction decomposition

$$\frac{h}{g} = \frac{h_1}{g_1} + \frac{h^*}{g^*},$$

with unique $h_1, h^* \in F[x]$ such that $\deg h_1 < \deg g_1$ and $\deg h^* < \deg g^*$. The first fraction has a squarefree denominator; it contributes to the logarithmic part of $\int (h/g)$.

Prove that d divides bd' and that $\gcd(bd'/d, b) = 1$. Thus there exist $s, t \in F[x]$ of smaller degree than b such that $s(bd'/d) + tb = h^*$. Use this to write

$$\frac{h^*}{g^*} = \left(\frac{u}{d}\right)' + \frac{v}{d},$$

with $u, v \in F[x]$ of smaller degree than d, and determine u and v.

(ii) Use (i) iteratively to split $\int (h/g)$ into its rational and logarithmic parts, and show that this takes $O(m\mathsf{M}(n)\log n)$ operations in F.

(iii) Analyze both the Hermite reduction as described in Section 22.2 and Mack's variant when using classical polynomial arithmetic, and compare your results.

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22.9 What is the leading coefficient of the resultant r in Theorem 22.8? Hint: Prove that it is the constant coefficient of $\operatorname{res}_x(ay-b',b)$.

22.10$^{\longrightarrow}$ Trace Algorithm 22.10 on computing the integral of

$$f = \frac{x^9}{x^7+3x^6-5x^5-23x^4-8x^3+40x^2+48x+16} \in \mathbb{Q}(x).$$

22.11 Let F be a field of characteristic zero and $a,b \in F[x]$ nonzero and coprime.

(i) Let $\gamma \in F$ and $p \in F[x]$ be an irreducible factor of $\gcd(b,a-\gamma b')$. Prove that $p^2 \nmid b$, and conclude that the gcd is squarefree.

(ii) Show that $\gcd(b,a-\gamma_1 b')$ and $\gcd(b,a-\gamma_2 b')$ are coprime if $\gamma_1,\gamma_2 \in F$ are distinct.

(iii) Consider the following variant of Algorithm 22.14.

ALGORITHM 22.22 Almkvist & Zeilberger's multiple of integration denominator.

Input: Relative prime polynomials $a,b \in F[x]$ with $b \neq 0$ monic.

Output: A monic polynomial $V \in F[x]$ such that for any coprime $u,v \in F[x]$, equation (5) implies that v divides V.

1. $R \longleftarrow \operatorname{res}_x(b,a-yb')$, $\quad d \longleftarrow \max\{i \in \mathbb{N} : i=0 \text{ or } R(i)=0\}$
   if $d=0$ **then return** 1
2. $a_0 \longleftarrow a$, $\quad b_0 \longleftarrow b$
   for $i=1,\ldots,d$ **do**
   $$H_i \longleftarrow \gcd(b_{i-1}, a_{i-1}-b'_{i-1}), \quad a_i \longleftarrow \frac{a_{i-1}-b'_{i-1}}{H_i}, \quad b_i \longleftarrow \frac{b_{i-1}}{H_i}$$
3. **return** $H_1 H_2^2 \cdots H_d^d$

Show that $H_i = \gcd(b,a-ib')$ for all i and conclude that the algorithm returns the same result as Algorithm 22.14.

(iv) Trace Algorithm 22.22 on the input from Example 22.16.

22.12 Let F be a field of characteristic zero and $r,s,t,U \in F[x]$ with $rU'-sU=t$ and $\gcd(r,s)=1$. Suppose that $\deg\gcd(r,t) \geq 1$, and reduce this differential equation for U to a differential equation for a proper divisor U^* of U, with coefficients r, s^*, and $t^* = t/\gcd(r,t)$.

22.13 Let F be a field of characteristic zero and $r,s,t \in F[x]$ such that r is nonzero and monic.

(i) Let $S = \{H \in F[x] : rH'-sH=0\}$ and $H_1,H_2 \in S \setminus \{0\}$. Show that there is a nonzero constant $c \in F$ such that $H_1 = cH_2$. Hint: Lemma 22.18.

(ii) Prove that either $S=\{0\}$ or there is a unique monic polynomial $H_0 \in S$, which has degree δ, where δ is as in Lemma 22.18, such that $S = \{cH_0 : c \in F\}$.

(iii) Suppose that the inhomogeneous equation (8) has a solution $U \in F[x]$. Prove that the set of all solutions to this equation is $\{U+H : H \in S\}$.

https://doi.org/10.1017/CBO9781139856065.028 Published online by Cambridge University Press

Summa cum laude.[1]

A sum to trick th' arithmetic.
Rudyard Kipling (1893)

Some merely took refuge in the mathematics, chains of difficult calculations using symbols as stepping stones on a march through fog.
James Gleick (1992)

As an algorist Euler has never been surpassed, and probably never even closely approached, unless perhaps by Jacobi. An algorist is a mathematician who devises "algorithms" (or "algorisms") for the solution of problems of special kinds [. . .] There is no uniform mode of procedure—algorists, like facile rhymesters, are born, not made. It is fashionable today to despise the 'mere algorist'; yet, when a truly great one like the Hindu Ramanujan arrives unexpectedly out of nowhere, even expert analysts hail him as a gift from Heaven [. . .] An algorist is a 'formalist' who loves beautiful formulas for their own sake.
Eric Temple Bell (1937)

Sometimes when studying his work I have wondered how much Ramanujan could have done if he had had MACSYMA or SCRATCHPAD or some other symbolic algebra package. More often I get the feeling that he was such a brilliant, clever, and intuitive computer himself that he really did not need them.
George E. Andrews (1986)

[1] Sum with praise ;-)

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